

MSc Quants Cluster
Risk Analysis: Group Coursework

**Measuring Market Risk:
VaR and Expected Shortfall Estimation and Backtesting
Analysis**

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Introduction

Value at Risk and Expected Shortfall are the primary regulatory measures of market risk under Basel III and IV, yet their accuracy depends critically on modelling assumptions about return distributions and volatility dynamics. The March 2020 COVID-19 crash, which produced an 8-standard-deviation loss in a single day for the portfolio studied here, and the prolonged 2022 technology selloff illustrate how quickly Gaussian assumptions can fail, potentially leaving institutions undercapitalised precisely when capital is most needed.

This report investigates whether standard parametric and non-parametric VaR/ES methods can adequately capture the tail risk of a concentrated equity portfolio exhibiting fat tails, negative skewness, and pronounced volatility clustering, as well as the tail risk of a bond instrument.

The analysis is carried out in four parts. Section 1 constructs an equally weighted portfolio of six large-cap US technology stocks (Apple, Microsoft, IBM, NVIDIA, Alphabet, Amazon) over January 2014 to December 2025 and evaluates five estimation methods, Gaussian parametric, Historical Simulation, Student-*t* Method of Moments, and rolling and expanding GARCH(1,1), using formal backtests of coverage, independence, and distributional fit. Section 2 assesses the statistical power of the Kupiec unconditional coverage test via Monte Carlo simulation, quantifying the test's ability to detect misspecification under a realistic GARCH alternative. Section 3 compares three portfolio allocation strategies, Risk Parity, Maximum Diversification, and Equal Weighting, in terms of risk contribution, out-of-sample performance, and VaR accuracy. Section 4 analyses interest rate risk for a 10-year coupon bond, comparing exact revaluation with delta-normal, delta-gamma, and Monte Carlo methods to evaluate the reliability of analytical approximations under large yield movements.

1. VaR and ES Estimation and Backtesting

1.1 Statistical Analysis of the Portfolio's Returns

An equally weighted portfolio of six large-cap US technology stocks (Apple, Microsoft, IBM, NVIDIA, Alphabet, Amazon) is constructed from daily adjusted closing prices sourced from Bloomberg over 1 January 2014 to 31 December 2025, yielding 2,997 log-return observations.

1.1.1 Descriptive Statistics

The portfolio has an annualised mean return of 24.66% and annualised volatility of 23.99% (daily SD 1.511%). Skewness of -0.337 indicates mild left asymmetry; excess kurtosis of 6.34 indicates substantially heavier tails than the normal, as confirmed by the distribution and Q-Q plot in Figure 2. The minimum daily return of -12.76% (16 March 2020) lies more than eight standard deviations from the mean, visible as an isolated spike in Figure 1. The empirical 1st percentile of -4.37% is 24% more extreme than the Gaussian prediction of -3.52% ($= 1.511\% \times 2.326$): downside risk is systematically understated by the Gaussian assumption. Full summary statistics are reported in Table 1.

Table 1: Descriptive Statistics — Equal-Weighted Portfolio (Jan 2014 – Dec 2025)

Statistic	Value
Observations	2,997
Annualised Mean Return	24.66%
Annualised Volatility	23.99%
Daily Standard Deviation	1.511%
Skewness	-0.337
Excess Kurtosis	6.34
Minimum (16 Mar 2020)	-12.76%
Maximum (9 Apr 2025)	$+11.32\%$
Empirical 1st Percentile	-4.37%
Gaussian 1st Percentile ($1.511\% \times 2.326$)	-3.52%
Tail Shortfall (empirical vs Gaussian)	24%
Maximum Drawdown	-40.02%

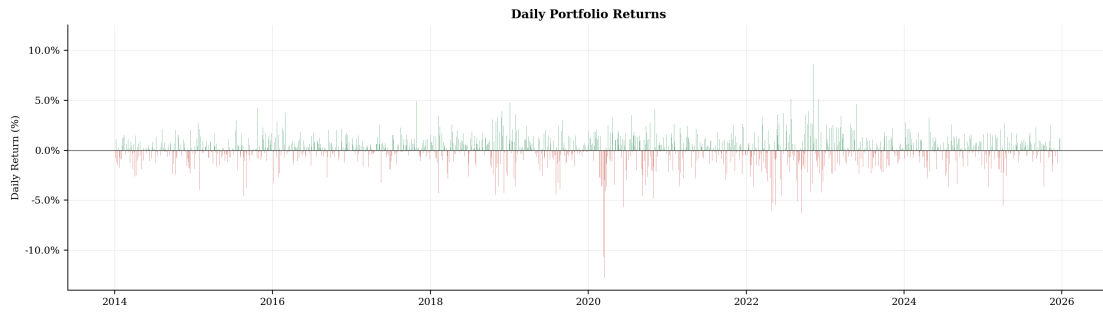


Figure 1: Daily portfolio log-returns (January 2014 – December 2025). Volatility is clearly time-varying, with pronounced clustering around the March 2020 crash and the 2022 selloff.

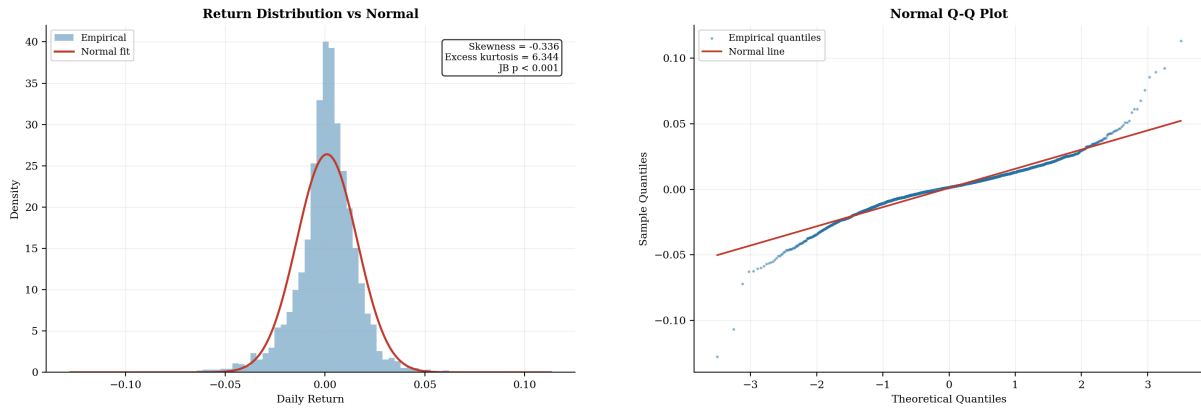


Figure 2: Left: return distribution with fitted normal overlay. Right: normal Q-Q plot. Both confirm heavy tails and mild negative skewness relative to the Gaussian benchmark, consistent with the excess kurtosis of 6.34 reported in Table 1.

1.1.2 Normality Tests

All four normality tests reject at $p \approx 0$, as reported in Table 2. An excess kurtosis of 6.34 is large enough that non-normality is not a sample-size artefact. The resulting 24% shortfall between the Gaussian and empirical 1st percentile motivates the fat-tailed alternatives in Section 1.2.

Table 2: Normality Test Results

Test	Null Hypothesis	Statistic	p -value	Decision
Jarque–Bera	Skewness = 0, excess kurtosis = 0	5,082.02	≈ 0	Reject
Kolmogorov–Smirnov	Returns $\sim \mathcal{N}(\hat{\mu}, \hat{\sigma}^2)$	0.0807	≈ 0	Reject
Shapiro–Wilk	Returns normally distributed	0.9364	≈ 0	Reject
Anderson–Darling	Returns normally distributed	36.56	≈ 0	Reject

1.1.3 Serial Dependence and Volatility Clustering

The Ljung-Box statistic on raw returns is 137.22 ($p \approx 0$), but individual autocorrelations of 0.05–0.08 explain under 1% of return variance; the conditional mean can be treated as approximately i.i.d. for VaR purposes. Evidence for time-varying volatility is far stronger: the Ljung-Box on squared returns is 2,160.45 ($p \approx 0$) with autocorrelations of 0.3–0.4 at short lags, as shown in Figure 3, and the ARCH-LM test (10 lags) yields 604.39 ($p \approx 0$). Full test results are reported in Table 3. Rolling volatility ranges from 15% in calm periods to above 50% during March 2020, a range no constant-volatility model can accommodate (Figure 4).

Table 3: Serial Dependence Test Results

Test	Series Tested	Statistic	p -value	Decision
Ljung–Box (20 lags)	Raw returns r_t	137.22	≈ 0	Reject i.i.d.
Ljung–Box (20 lags)	Squared returns r_t^2	2,160.45	≈ 0	Reject no ARCH
ARCH-LM (10 lags)	Raw returns r_t	604.39	≈ 0	Reject no ARCH

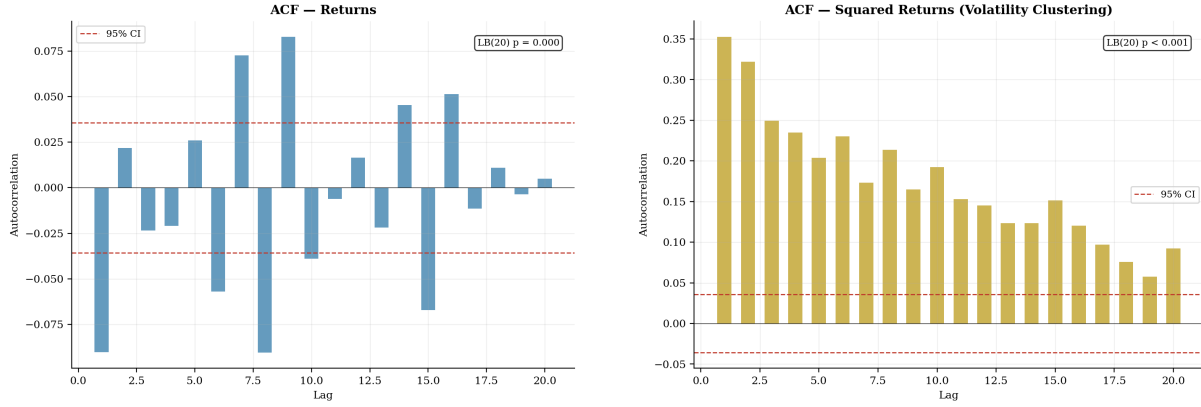


Figure 3: ACF of raw returns (left) and squared returns (right). Autocorrelations of squared returns are large and persistent at all lags, confirming strong volatility clustering as quantified by the Ljung–Box and ARCH-LM statistics in Table 3

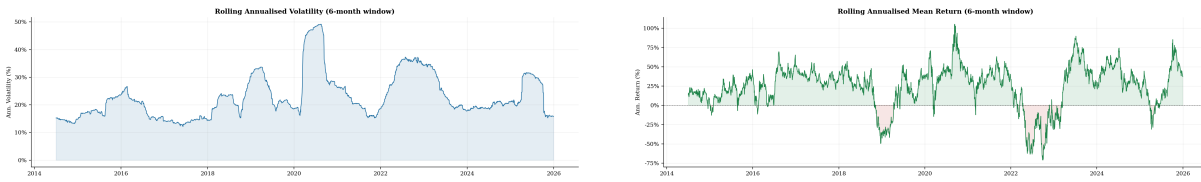


Figure 4: Rolling 6-month annualised volatility (left) and mean return (right). Volatility ranges from approximately 15% in calm periods to above 50% during March 2020, underscoring the time-variation that motivates the GARCH models in Section 1.2.

Three features bear directly on model selection: excess kurtosis of 6.34 (fat tails understating 99% VaR by 24%); skewness of -0.337 (heavier left tail); and strong, persistent volatility clustering. A Gaussian constant-volatility model misses all three, producing systematic backtesting failures across every parametric method tested (Section 1.3).

1.2 Rolling VaR and Expected Shortfall Estimation

1.2.1 Estimation Framework

VaR and ES are estimated at the 99% confidence level for a 1-day horizon using a rolling 6-month window of 125 trading days. At each date t , the window covers the 125 observations ending at $t - 1$, and the resulting parameters forecast risk on day t . Estimation starts on 1 July 2014, giving 2,873 daily estimates through December 2025. Five methods are implemented, summarised in Table 4 below; formulas and key properties of these methods are detailed in the subsequent subsections.

Table 4: Overview of the five VaR/ES estimation methods

Method	Distribution Assumption	Key Limitation
Gaussian Parametric	Normal, constant volatility	Fat tails and clustering ignored
Historical Simulation	Non-parametric (empirical)	Ghost effect from rolling window
Student- t MoM	Student- t , constant volatility	No volatility dynamics; fallback to Gaussian if $\hat{K} \leq 0$
GARCH(1,1), Rolling	Normal innovations, time-varying σ^2	Unstable parameters on short window
GARCH(1,1), Expanding	Normal innovations, time-varying σ^2	Gaussian innovations still misspecified

1.2.2 Method 1: Gaussian Parametric

Rolling mean $\hat{\mu}$ and standard deviation $\hat{\sigma}$ are estimated on each 125-day window. VaR and ES are:

$$\text{VaR}_\alpha = -(\hat{\mu} + z_\alpha \hat{\sigma}), \quad \text{ES}_\alpha = -\left(\hat{\mu} - \hat{\sigma} \frac{\varphi(z_\alpha)}{1 - \alpha}\right) \quad (1)$$

where $z_\alpha = \Phi^{-1}(0.01) = -2.326$ and φ is the standard normal density. This approach is misspecified given the fat tails and volatility clustering documented in Section 1.1.

1.2.3 Method 2: Historical Simulation

HS makes no distributional assumption. VaR is the empirical $(1 - \alpha)$ -quantile of the sorted 125-day window using linear interpolation. With the window sorted as $r_{(1)} \leq \dots \leq r_{(T)}$ and $k = \lfloor (1 - \alpha)T \rfloor$:

$$\text{VaR}_\alpha = -[(1 - q)r_{(k)} + q r_{(k+1)}], \quad \text{ES}_\alpha = -\frac{1}{T(1 - \alpha)} \left[\sum_{j=1}^{k-1} r_{(j)} + q r_{(k)} \right] \quad (2)$$

where $q = (1 - \alpha)T - k$. The ghost effect is the main drawback: a large loss drops out of the window abruptly after 125 days, causing a step change in VaR unrelated to current market conditions.

1.2.4 Method 3: Student- t Method of Moments

The degrees-of-freedom parameter ν is estimated by matching the theoretical excess kurtosis $K(\nu) = 6/(\nu - 4)$ to the sample excess kurtosis $\hat{K}$, giving $\hat{\nu} = 4 + 6/\hat{K}$. This requires $\hat{K} > 0$; if not, the Gaussian fallback applies. The scaled volatility is $\hat{\sigma}_t = \hat{\sigma} \sqrt{(\hat{\nu} - 2)/\hat{\nu}}$, and:

$$\text{VaR}_\alpha = -(\hat{\mu} + \hat{\sigma}_t t_\alpha(\hat{\nu})), \quad \text{ES}_\alpha = -\left(\hat{\mu} - \hat{\sigma}_t \frac{f(t_\alpha)}{1 - \alpha} \cdot \frac{\hat{\nu} + t_\alpha^2}{\hat{\nu} - 1}\right) \quad (3)$$

where $t_\alpha(\hat{\nu})$ is the lower α -quantile of the standard Student- t and $f(t_\alpha)$ its density. Of the 2,873 windows, 235 (8.2%) produce non-positive excess kurtosis, concentrated in 2022–2024 when the Fed tightening cycle produced a sustained directional drawdown that compressed rolling kurtosis, as illustrated in Figure 5. On those days the Gaussian fallback applies, giving 2,638 Student- t estimates.

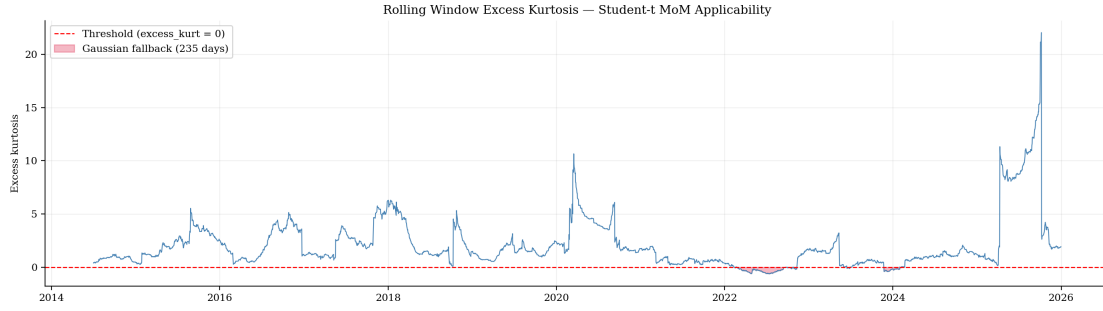


Figure 5: Rolling 125-day excess kurtosis. Windows producing $\hat{K} \leq 0$ (shaded) revert to the Gaussian fallback; 235 of 2,873 windows (8.2%) are affected, concentrated in 2022–2024.

1.2.5 Method 4: GARCH(1,1), Rolling Window

GARCH(1,1) models time-varying conditional variance:

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (4)$$

Parameters (ω, α, β) are estimated by maximum likelihood on each 125-day window. To reduce sensitivity to starting values, four initialisations are used:

$$(\alpha_0, \beta_0) \in \{(0.05, 0.80), (0.10, 0.85), (0.15, 0.75), (0.20, 0.70)\}$$

with $\omega_0 = \hat{\sigma}^2(1 - \alpha_0 - \beta_0)$ to respect the unconditional variance identity. The solution with the highest log-likelihood is retained. The one-step forecast $\sigma_{t+1}^2 = \hat{\omega} + \hat{\alpha} r_t^2 + \hat{\beta} \sigma_t^2$ is substituted into the Gaussian VaR and ES formulas. Mean persistence across windows is $\hat{\alpha} + \hat{\beta} = 0.820$, but 100 windows produced persistence below 0.1 and 131 above 0.99, reflecting the difficulty of identifying GARCH parameters on 125 observations; the resulting VaR instability is visible in Figure 6.

1.2.6 Method 5: GARCH(1,1), Expanding Window

The expanding window variant uses the same GARCH(1,1) specification but grows the estimation window from January 2014. By end of sample the window covers approximately 2,900 observations, considerably reducing parameter instability. Parameters are re-estimated every 21 trading days and held fixed between updates; the variance recursion is updated daily with the most recently fitted parameters.

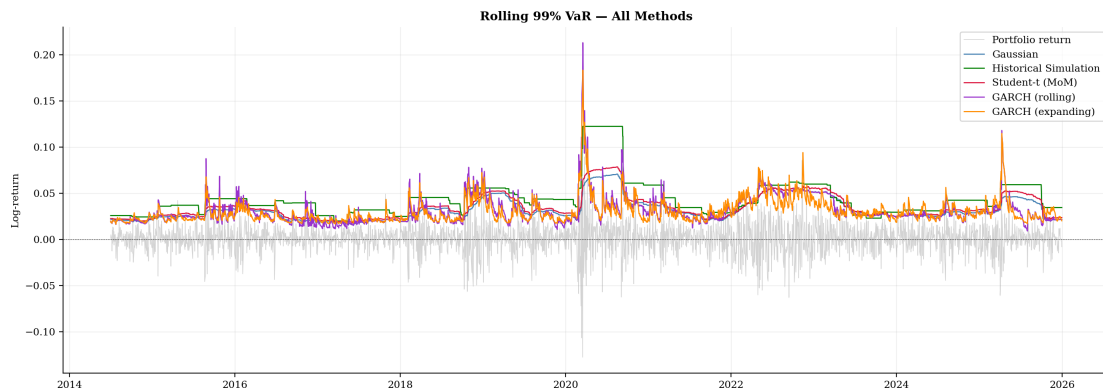


Figure 6: Rolling 99% one-day VaR across all five methods (July 2014 – December 2025). HS VaR exhibits the characteristic ghost-effect plateaus; GARCH VaR spikes sharply during March 2020.

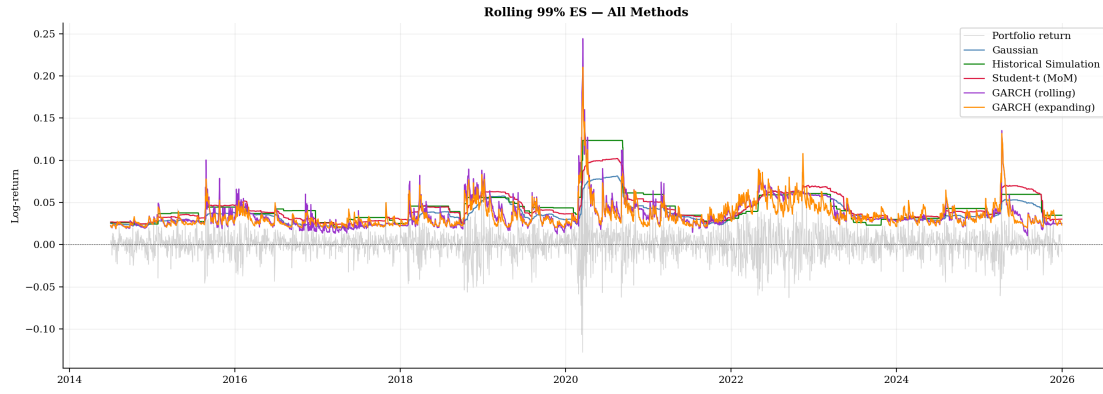


Figure 7: Rolling 99% Expected Shortfall across all five methods. ES tracks VaR closely in level but is uniformly higher, reflecting the conditional mean of tail losses.

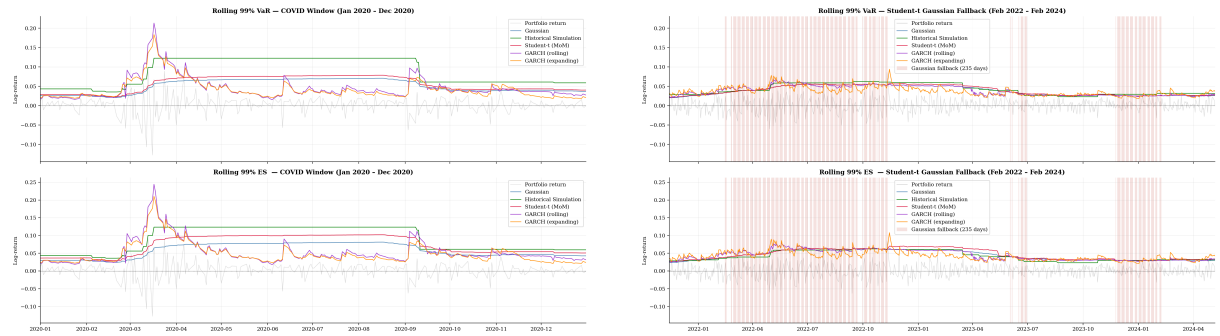


Figure 8: Left: VaR and ES during the COVID window (January–December 2020). Rolling GARCH VaR spikes above 20% as MLE over-weights extreme observations; expanding GARCH responds more gradually; HS VaR remains elevated for approximately six months after the crash before dropping in September 2020. Right: Student- t fallback period (2022–2024), showing days when the model reverts to Gaussian.

Figures 6 and 7 show the full time series of estimated VaR and ES across all five methods. The divergence between methods is most pronounced during stress episodes: GARCH rolling VaR spikes sharply in March 2020 before contracting as the short window rolls past the crash, while HS VaR plateaus for approximately six months after the crash before dropping abruptly. The expanding GARCH and Gaussian estimates respond more gradually, reflecting their broader estimation windows. Figure 8 isolates the COVID window and the 2022–2024 Student- t fallback period, where the sustained directional selloff compressed rolling kurtosis below zero on 235 days, forcing reversion to the Gaussian.

1.3 Violations and Backtesting

VaR violations and formal backtests are closely related: violations are the raw material from which all backtesting statistics are computed, and their frequency and temporal pattern jointly determine whether a model is rejected. This subsection first defines violations and reports counts, then introduces the testing framework, and finally synthesises results and model-level conclusions.

1.3.1 Violation Definition and Counts

A violation on day t occurs when the realised return falls below the predicted VaR:

$$I_t = \mathbf{1}\{r_t < -\text{VaR}_{\alpha,t-1}\} \quad (5)$$

Under a correctly specified model, $\mathbb{E}[I_t] = 1\%$ and violations are serially independent. Table 5 reports counts and rates for all five methods over the 2,873-day evaluation period. Historical Simulation is the only method close to the 1% target (1.11%), while all parametric methods substantially exceed it, ranging from 2.37% to 2.89%. The uniform overshoot reflects the fat tails and volatility clustering documented in Section 1.1: the March 2020 crash and the 2022 selloff concentrated violations in ways that a 125-day estimation window cannot anticipate. The temporal distribution of violations is illustrated in Figure 9.

Table 5: VaR Violation Counts and Rates (2,873 evaluation days, 99% confidence level)

Method	Violations	Rate (%)	vs. 1% target
Gaussian Parametric	83	2.89	+189%
Historical Simulation	32	1.11	+11%
Student- t MoM	68	2.37	+137%
GARCH(1,1) Rolling	76	2.65	+165%
GARCH(1,1) Expanding	68	2.37	+137%
Expected (correctly spec.)	29	1.00	—

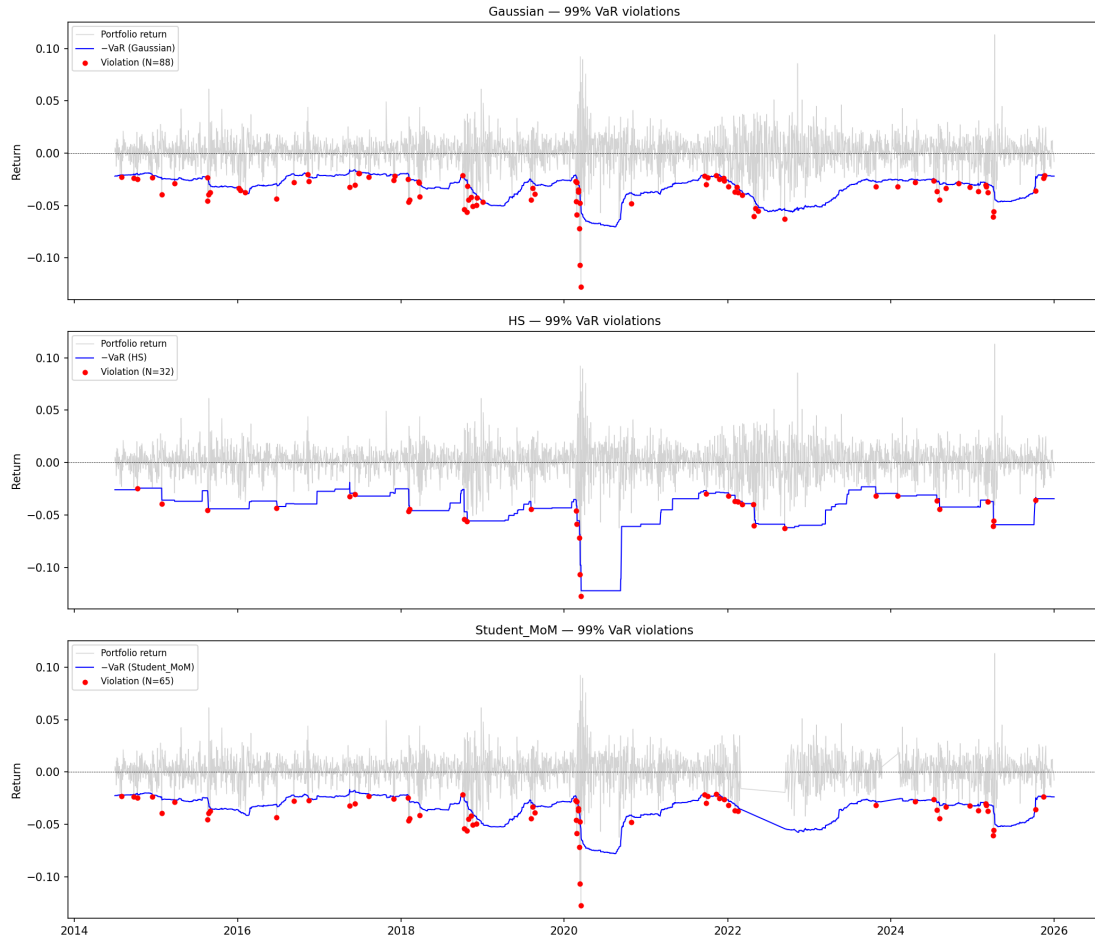


Figure 9: VaR violations across all five methods. Violations are concentrated in March 2020 and the 2022 selloff for all parametric models; HS shows fewer violations due to the ghost effect keeping VaR elevated, as also reflected in the counts in Table 5.

1.3.2 Testing Framework

Five tests are applied at the 5% significance level throughout, as summarised in Table 6.

Table 6: Backtesting tests applied

Test	What it assesses	Statistic	HS?
Kupiec UC	Correct unconditional violation frequency: $\mathbb{E}[I_t] = 1\%$	$LR_{uc} \sim \chi^2(1)$	Yes
Christoffersen IND	Serial independence via 2×2 transition matrix; detects clustering	$LR_{ind} \sim \chi^2(1)$	Yes
Conditional Coverage (CC)	Joint frequency and independence; primary adequacy criterion	$LR_{cc} \sim \chi^2(2)$	Yes
KS (distributional)	PITs uniform: $u_t = F_t(r_t/\hat{\sigma}_t) \sim U(0, 1)$	KS statistic	No
Berkowitz (2001)	Normalised PITs $z_t = \Phi^{-1}(u_t) \sim \mathcal{N}(0, 1)$ i.i.d.; tests mean, variance, AR(1)	$LR_{Berk} \sim \chi^2(3)$	No

For the Student- t model, the PIT uses the time-varying Student- t CDF at $\hat{\nu}_t$, reverting to Gaussian on fallback days. Neither distributional test applies to Historical Simulation.

1.3.3 Results

Table 7 summarises the pass/fail outcome for all five tests and all five models.

Table 7: Backtesting Results — All Methods (5% significance level; ✓ = pass, ✗ = fail)

Method	UC	IND	CC	KS	Berkowitz
Gaussian Parametric	✗ ($p \approx 0$)	✗ ($p = 0.001$)	✗ ($p \approx 0$)	✗ ($p \approx 0$)	✗ ($p \approx 0$)
Historical Simulation	✓ ($p = 0.55$)	✓ ($p = 0.370$)	✓ ($p = 0.57$)	N/A	N/A
Student- t MoM	✗ ($p \approx 0$)	✗ ($p = 0.006$)	✗ ($p \approx 0$)	✗ ($p \approx 0$)	✗ ($p \approx 0$)
GARCH(1,1) Rolling	✗ ($p \approx 0$)	✓ ($p = 0.066$)	✗ ($p \approx 0$)	✗ ($p \approx 0$)	✗ ($p \approx 0$)
GARCH(1,1) Expanding	✗ ($p \approx 0$)	✓ ($p = 0.102$)	✗ ($p \approx 0$)	✗ ($p \approx 0$)	✗ ($p \approx 0$)

Unconditional Coverage. The Kupiec test rejects all parametric models ($p \approx 0$); violation rates of 2.37–2.89% (Table 5) are incompatible with the 1% nominal level (LR_{uc} : Gaussian 68.61, Student- t 39.18, GARCH rolling 54.11, GARCH expanding 39.18). Historical Simulation is not rejected ($LR_{uc} = 0.36$, $p = 0.55$), though its low count could reflect the ghost effect rather than good calibration: a sustained elevated VaR mechanically reduces violations during the recovery period following a crash.

Independence. The IND test rejects the Gaussian ($p = 0.001$, $LR_{ind} = 11.73$) and Student- t ($p = 0.006$, $LR_{ind} = 7.61$): their violations cluster. Both GARCH variants pass ($p = 0.066$ rolling; $p = 0.102$ expanding), as does HS ($p = 0.370$). The dynamic variance recursion reduces clustering, though not the overall violation count, leaving the UC test unsatisfied.

Conditional Coverage. All parametric models fail CC ($p \approx 0$). HS passes ($LR_{cc} = 1.13$, $p = 0.57$), passing both sub-tests simultaneously: its violation count is close to the 1% target and violations are serially independent. The CC test is the primary criterion of model adequacy; by this standard, only HS is not rejected — though for the wrong reasons (the ghost effect, not accurate risk estimation).

Distributional Tests. All parametric models fail both KS and Berkowitz ($p = 0$ throughout; Berkowitz LR statistics in the thousands). The PITs are far from uniform, indicating broad distributional misspecification beyond tail frequency alone. That the GARCH models fail these tests despite their dynamic volatility adjustment points to the Gaussian innovation assumption as the primary source of failure.

1.3.4 Model Assessment

Gaussian. Misspecified on all three dimensions. The 24% underestimation of the empirical 1st percentile translates into a violation rate three times the nominal level. Violations also cluster (IND rejected, $p = 0.001$) because the rolling sample variance is slow to respond to volatility shocks. All distributional tests reject decisively.

Historical Simulation. HS passes all applicable tests, but its low violation count is a product of the ghost effect. The large losses of 2020 and 2022 remain in the 125-day window for months after they occurred, keeping VaR artificially elevated during recovery periods. Figure 8 shows HS VaR remaining elevated for approximately six months after the COVID crash before dropping sharply in September 2020. Passing backtesting through systematic over-reservation is not the same as accurate risk measurement.

Student- t MoM. The Student- t addresses fat tails but not volatility clustering. Its violation rate of 2.37% is below the Gaussian but still more than twice the nominal level, and violations cluster (IND $p = 0.006$). A further limitation is that the MoM estimator requires positive excess kurtosis, which fails on 235 days (8.2%) during 2022–2024 when the sustained directional selloff compressed rolling kurtosis below zero, forcing reversion to Gaussian as shown in the right panel of Figure 8. On those days the model degrades to Gaussian.

GARCH(1,1), Rolling vs Expanding. GARCH directly targets the dominant feature of the return series. Both variants pass the IND test — the only parametric models to do so — showing that the dynamic variance recursion reduces violation clustering. The UC and CC tests still fail, because the Gaussian innovation assumption leaves the conditional tails too thin: standardised residuals $r_t/\hat{\sigma}_t$ are not normally distributed, as the KS and Berkowitz rejections confirm. The expanding window improves on the rolling variant: across its estimation points it produces no degenerate fits (mean persistence 0.917, concentrated in $[0.82, 0.97]$), compared to the rolling window’s 100 near-zero and 131 near-unit persistence windows on 125-observation samples. The violation rate falls from 2.65% to 2.37%. Figure 8 shows the difference most clearly during the COVID crash, where rolling GARCH VaR spikes above 20% as MLE over-weights extreme observations in a short window, while the expanding GARCH responds more gradually.

1.4 Conclusion

The portfolio exhibits excess kurtosis of 6.34, mild negative skewness, and strong persistent volatility clustering, three features that no single method in this analysis addresses simultaneously. Rolling VaR estimates diverge sharply during stress episodes: GARCH rolling spikes above 20% in March 2020 before contracting as the short window rolls past the crash, while HS VaR plateaus for six months due to the ghost effect, and Gaussian estimates respond too slowly throughout. In backtesting, Historical Simulation is the only method not formally rejected, but its low violation count reflects the ghost effect rather than accurate calibration. All parametric models produce violation rates of 2.37–2.89% against the 1% target, with the GARCH variants uniquely passing the independence test by capturing clustering but still failing on tail thickness due to Gaussian innovations. These results are compounded by structural constraints: fixed equal weights ignore drift and rebalancing, the top-down approach discards constituent-level dynamics, and the 125-day window is too short for reliable GARCH parameter identification. The consistent finding is that fat tails and volatility clustering must be addressed jointly, a task that requires, at minimum, a fat-tailed innovation distribution within a dynamic volatility framework.

2. Power of the Kupiec Test

This section estimates the power of the Kupiec unconditional coverage test via Monte Carlo simulation, quantifying how reliably the test detects a misspecified Gaussian constant-volatility VaR model when the true process is a GARCH(1,1). Both models are calibrated to the portfolio return series from Section 1.1.

2.1 Statistical Framework

The power of a test is defined as one minus the probability of a Type 2 error: the probability of correctly rejecting H_0 when it is false. The Kupiec (1995) Proportion-of-Failures test assesses whether the observed violation count N is consistent with the nominal tail probability p . Under a correctly specified model, $N \sim \text{Binomial}(T, p)$. The likelihood ratio statistic is:

$$\text{LR}_{\text{uc}} = -2 \left[N \ln \frac{p}{\hat{p}} + (T - N) \ln \frac{1-p}{1-\hat{p}} \right] \sim \chi^2(1) \text{ under } H_0 \quad (6)$$

where $\hat{p} = N/T$. H_0 is rejected at the 5% level when $\text{LR}_{\text{uc}} > 3.84$. Boundary cases $N = 0$ or $N = T$ are handled using the exact two-sided binomial p -value.

H_0 : returns are i.i.d. $\mathcal{N}(0, \sigma^2)$, with σ estimated as the full-sample standard deviation, giving $\text{VaR} = \sigma\Phi^{-1}(0.99)$. H_1 : returns follow GARCH(1,1) with $r_t = \sigma_t\varepsilon_t$, $\varepsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$.

2.2 Monte Carlo Procedure

Both models are fitted to the full $T = 2,997$ daily portfolio log-returns. Under H_0 , $\sigma = 1.5113\%$ /day. Under H_1 , GARCH(1,1) MLE yields $\hat{\omega} = 8.53 \times 10^{-6}$, $\hat{\alpha} = 0.1248$, $\hat{\beta} = 0.8360$, persistence = 0.9608, unconditional $\sigma = 1.4759\%$ /day. The unconditional volatility is nearly identical under both hypotheses, confirming that any power differences reflect clustering structure, not a shift in the marginal distribution. The VaR threshold is computed from H_0 and held fixed throughout, replicating a setting in which a risk manager applies a Gaussian model while the true process is GARCH. For each of $M = 10,000$ replications, paths of length T are drawn from both H_0 and H_1 and the Kupiec test is applied. Three dimensions are varied in turn, as described in Table 8.

Table 8: Simulation design: each row varies one dimension holding the others fixed

Dimension	Values varied	Fixed parameters
Sample size T	250, 500, 750, 1,000, 1,500, 2,000, 2,500, 3,000	$p = 1\%$, fitted GARCH
Confidence level ($1 - p$)	90%, 95%, 97.5%, 99%, 99.5%, 99.9%	$T = 2,997$, fitted GARCH
GARCH persistence $\alpha + \beta$	0.70, 0.80, 0.85, 0.90, 0.93, 0.96, 0.98	$T = 2,997$, $p = 1\%$; unconditional variance fixed at σ^2

2.3 Results

The Type 1 error rate stays close to 5% across all experiments, confirming that the $\chi^2(1)$ approximation is well-calibrated. Results are summarised in Table 9 and Figure 10.

Table 9: Summary of Kupiec power simulation results

Panel	Dimension	Key Finding	Power Range
(a)	Sample size T	Power rises with T but non-monotonically; high persistence causes regime effects at intermediate T	13% ($T = 250$) to 39% ($T = 3,000$)
(b)	Confidence level p	U-shaped: minimum near 97.5–99%; 99% sits near the trough	36–40% at 97.5–99%; 75% at 90%
(c)	GARCH persistence $\alpha + \beta$	Monotone increase; high persistence prolongs volatility regimes, clustering violations	7% ($\alpha + \beta = 0.70$) to 54% ($\alpha + \beta = 0.98$)
(d)	Violation distributions	H_1 mean 23% higher than H_0 ; substantial overlap explains moderate power	H_0 : $\mu = 30.0$, $\sigma = 5.4$; H_1 : $\mu = 36.8$, $\sigma = 12.7$

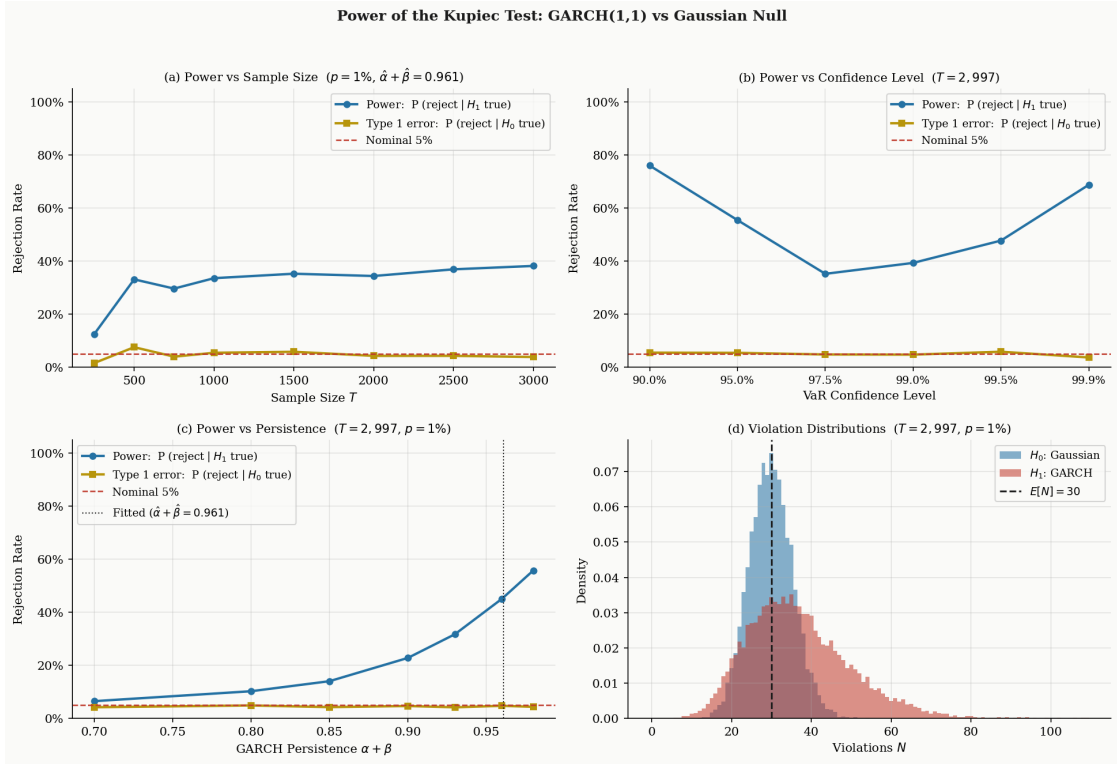


Figure 10: Power of the Kupiec unconditional coverage test across four dimensions: (a) sample size T , (b) confidence level, (c) GARCH persistence $\alpha + \beta$, (d) violation count distributions under H_0 and H_1 . Type 1 error rate (dashed) remains close to the nominal 5% throughout.

Power vs sample size (panel a). As shown in panel (a) of Figure 10, power rises from 13% at $T = 250$ to approximately 39% at $T = 3,000$, though the increase is not monotone: it jumps to 34% at $T = 500$, dips modestly around $T = 750$ – $1,000$, then climbs gradually. With persistence 0.9608, volatility cycles can last hundreds of days; at $T = 500$ a single high-volatility episode tends to dominate the path and generate a burst of violations, whereas at $T = 750$ – $1,000$ the path averages over more varied regimes, temporarily reducing power. At $T = 250$, the expected violation count is only 2.5, leaving the test with very limited discriminatory power.

Power vs confidence level (panel b). Panel (b) of Figure 10 shows that power follows a U-shaped pattern: high at 90% (75%), falling to a minimum around 97.5–99% (36–40%), then rising at 99.5% and 99.9%. At 90%, the expected violation count under H_0 is already 300 over this sample, giving the test ample observations to detect the excess generated by GARCH. At 99.9%, the expected count is only approximately 3, but a GARCH path that produces even a small cluster of extra violations generates a very large LR statistic, so detection is still relatively powerful. The minimum occurs around 97.5–99% because the expected violation count under H_0 is small enough (around 30 at 99%) that the two distributions overlap substantially, yet not so extreme that a single cluster dominates. The 99% confidence level used in Sections 1.2–1.3 sits near the minimum of this curve, with power of approximately 38%.

Power vs GARCH persistence (panel c). Panel (c) of Figure 10 shows that, holding the unconditional variance fixed, power increases monotonically with $\alpha + \beta$. At low persistence ($\alpha + \beta = 0.70$), shocks dissipate within days and the process is nearly indistinguishable from i.i.d., giving power of only 7%. At the fitted persistence of 0.9608, power is 45%; at 0.98 it reaches 54%.

Violation distributions (panel d). Panel (d) of Figure 10 shows the distribution of the violation count N across 10,000 replications under both hypotheses. Under H_0 , N follows $\text{Binomial}(T = 2,997, p = 0.01)$, giving a mean of 30.0 and standard deviation of 5.4 — a tight, well-behaved distribution. Under H_1 , the mean rises to 36.8 (+23%) because GARCH clustering concentrates losses in ways that breach the fixed Gaussian VaR threshold more often. However, the standard deviation explodes to 12.7, more than twice the H_0 value, because GARCH paths are highly variable: some sit in calm regimes and produce very few violations, others hit sustained high-volatility periods and produce many. The two

distributions therefore overlap substantially, which is why power at $T = 2,997$ and $p = 1\%$ is moderate despite the mean shift.

2.4 Conclusion for Backtesting Results

Power of approximately 38% at the relevant horizon and confidence level means that the Kupiec test has a substantial Type 2 error rate: under the GARCH alternative, roughly 62% of replications would not trigger rejection. All parametric models are nonetheless rejected in Section 1.3, with violation rates of 2.37–2.89%, constituting strong evidence of misspecification since the test tends to err on the side of acceptance in this setting. The excess dispersion of N under GARCH also helps interpret the Christoffersen independence test: violations arriving in clusters inflate both the total count and the transition probability $P(I_t = 1 \mid I_{t-1} = 1)$.

3. The Risk Parity Portfolio

This section analyses the performance of three portfolio allocation strategies using the dataset introduced in Section 1. The first half of the dataset (1,498 observations) determines portfolio weights; the second half (1,499 observations) is reserved for out-of-sample performance evaluation. The three strategies are Risk Parity (RP), Maximum Diversification (MD), and Equal Weighting (EW).

The **Risk Parity** portfolio allocates capital such that each asset contributes equally to overall portfolio risk, minimizing the standard deviation of Component VaRs (CVaRs). Two approaches are used: a parametric approach, assuming a Gaussian distribution with the sample covariance matrix, and a non-parametric approach relying on the empirical distribution of historical returns.

The **Maximum Diversification** portfolio maximizes the diversification ratio:

$$D(w) = \frac{w^\top \sigma}{\sqrt{w^\top \Sigma w}} \quad (7)$$

subject to $w^\top \mathbf{1} = 1$, $w \geq 0$, where σ is the vector of individual asset volatilities and Σ is the covariance matrix. The MD portfolio equalizes each asset’s correlation with the portfolio return.

The **Equal-Weight** portfolio assigns $w_i = 1/N$ to all assets.

3.1 Portfolio Weights and Component VaR Analysis

Table 10 reports the portfolio weights from the in-sample optimisation. The Risk Parity weights are similar under both methods, though not identical, since the parametric approach assumes normality whereas the non-parametric method relies on the empirical distribution. The Maximum Diversification portfolio exhibits a more concentrated allocation: IBM receives the largest weight (34.66%) because it has relatively high individual volatility but low correlation with the other technology stocks, contributing strongly to diversification; MSFT receives the smallest weight (3.25%).

Table 10: Portfolio weights for each allocation strategy

Stock	RP (Parametric)	RP (NonParam)	Max Diversification	Equal Weight
AAPL	0.1762	0.1839	0.1953	0.1667
MSFT	0.1725	0.1901	0.0325	0.1667
IBM	0.2241	0.1899	0.3466	0.1667
NVDA	0.1181	0.1233	0.1562	0.1667
GOOGL	0.1674	0.1685	0.0996	0.1667
AMZN	0.1418	0.1441	0.1698	0.1667

The parametric RP portfolio achieves perfect equalization of CVaR contributions at 16.67% per asset by construction (Table 11). The non-parametric RP (Table 12) produces contributions ranging from 15.89% to 17.59%; small deviations from equality arise because the non-parametric method relies on finite-window CVaR approximations.

Table 11: Component VaR for Risk Parity Portfolio (Parametric)

Stock	Weight	MPVaR	CVaR	CVaR Contribution (%)
AAPL	0.1762	0.0176	0.0031	16.67
MSFT	0.1725	0.0180	0.0031	16.67
IBM	0.2241	0.0139	0.0031	16.67
NVDA	0.1181	0.0263	0.0031	16.67
GOOGL	0.1674	0.0186	0.0031	16.67
AMZN	0.1418	0.0219	0.0031	16.67

Table 12: Component VaR for Risk Parity Portfolio (Non-Parametric)

Stock	Weight	MPVaR	CVaR	CVaR Contribution (%)
AAPL	0.1839	0.0167	0.0031	15.89
MSFT	0.1901	0.0172	0.0033	16.92
IBM	0.1899	0.0166	0.0032	16.36
NVDA	0.1233	0.0259	0.0032	16.55
GOOGL	0.1685	0.0191	0.0032	16.69
AMZN	0.1441	0.0236	0.0034	17.59

The MD portfolio (Table 13) exhibits a very different distribution of risk contributions, with IBM dominating at 27.98% and MSFT contributing only 2.75%. Table 14 shows that the correlation between each asset and the Maximum Diversification portfolio is approximately constant. This is consistent with the theoretical property of the MD portfolio, which equalises the correlation of each asset with the portfolio return.

Table 13: Component VaR for Maximum Diversification Portfolio

Stock	Weight	MPVaR	CVaR	CVaR Contribution (%)
AAPL	0.1953	0.0170	0.0030	18.08
MSFT	0.0325	0.0160	0.0005	2.75
IBM	0.3466	0.0150	0.0050	27.98
NVDA	0.1562	0.0280	0.0040	23.00
GOOGL	0.0996	0.0170	0.0020	8.96
AMZN	0.1698	0.0210	0.0040	19.22

Table 14: Correlation of each stock with Maximum Diversification Portfolio

Stock	Correlation with MD portfolio
AAPL	0.7244
MSFT	0.7244
IBM	0.7244
NVDA	0.7244
GOOGL	0.7244
AMZN	0.7244

The EW portfolio (Table 15) assigns identical weights but risk contributions differ substantially: NVDA contributes 24.35% and IBM only 10.91%, reflecting differences in volatility and correlation. Equal capital allocation does not imply equal risk allocation.

Table 15: Component VaR for Equally-Weighted Portfolio

Stock	Weight	MPaR	CPaR	CPaR Contribution (%)
AAPL	0.1667	0.0170	0.0030	14.76
MSFT	0.1667	0.0180	0.0030	15.14
IBM	0.1667	0.0130	0.0020	10.91
NVDA	0.1667	0.0290	0.0050	24.35
GOOGL	0.1667	0.0180	0.0030	15.74
AMZN	0.1667	0.0220	0.0040	19.10

3.2 Out-of-Sample Performance

Table 16 summarises the key performance metrics for all four strategies over the out-of-sample period.

Table 16: Performance metrics across portfolio strategies (out-of-sample)

Portfolio	Sharpe Ratio	Max Drawdown	Terminal Wealth	Cumulative Return
RP (Parametric)	0.958	−32.80%	4.43	342.7%
RP (NonParametric)	0.951	−34.04%	4.50	349.7%
Max Diversification	1.000	−30.88%	4.66	365.6%
Equal Weighting	0.970	−36.35%	4.93	392.9%

The Equally Weighted portfolio achieves the highest cumulative return (392.9%) and terminal wealth (4.93) when evaluated on the *second half of the dataset*, which serves as the out-of-sample period for performance assessment (Table 16). This is explained by NVDA’s weight: EW assigns the full 1/6 allocation to NVDA, while RP downweights it as the most volatile asset and MD allocates 15.62%. NVDA’s extraordinary appreciation over the period therefore benefits EW disproportionately. However, EW also suffers the largest maximum drawdown (−36.35%), reflecting the consequence of ignoring asset-specific risk characteristics.

Both RP portfolios achieve lower drawdowns than EW, reflecting their balanced risk allocation. The Maximum Diversification portfolio delivers the lowest drawdown (−30.88%) and the highest Sharpe ratio (1.000), making it the strongest risk-adjusted performer. The near-identical performance of the two RP variants is unsurprising: pairwise correlations among the six technology stocks range from 0.32 to 0.63. Risk Parity converges exactly to inverse-volatility weighting when all pairwise correlations are equal; when correlations are high and relatively uniform across assets, as here, the correlation structure has limited effect on the optimisation and weights are driven primarily by individual volatilities. The result is a weighting scheme close to equal-weight. The cumulative wealth trajectories in Figure 11 illustrate these differences clearly.

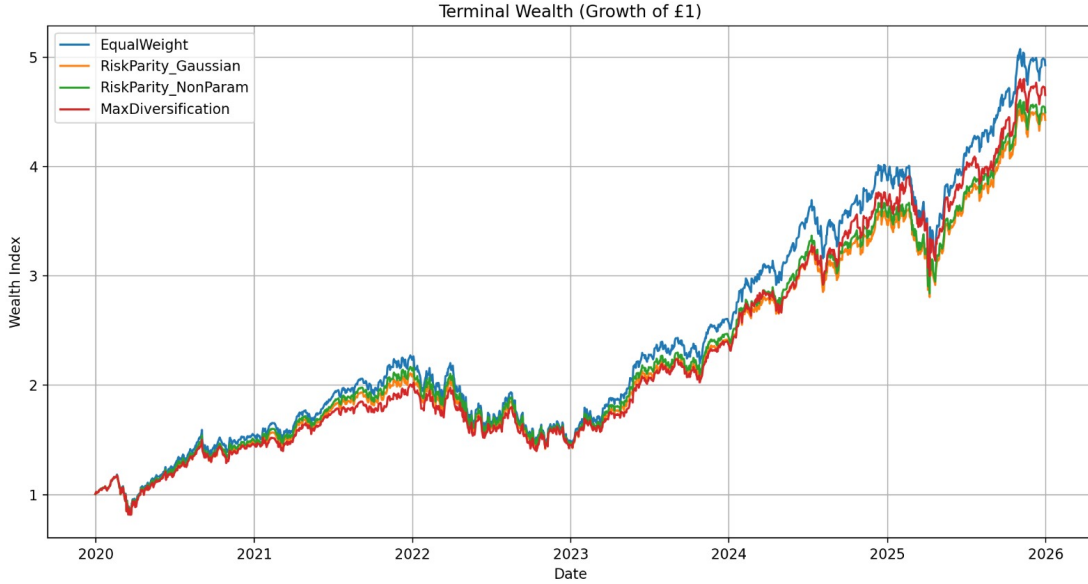


Figure 11: Terminal wealth path across portfolio strategies (out-of-sample period, growth of £1). The cumulative wealth index is $W_t = \exp\left(\sum_{s=1}^t r_{p,s}\right)$. The EW portfolio leads consistently from 2024 onward, driven by NVIDIA's AI-rally, while the MD portfolio achieves the lowest drawdown at -30.88% (Table 16).

3.3 VaR Backtesting

VaR violations are computed using a rolling window of 125 trading days at the 95% confidence level ($\alpha = 5\%$), as specified by the question. This differs from the 99% level used in Sections 1.1–1.3; the higher expected violation rate of 5% means formal backtesting tests would require fewer observations to achieve adequate power.

Table 17: Gaussian parametric VaR violations in the out-of-sample period

Portfolio	Observations	Violations	Expected (%)	Observed (%)
RP (Parametric)	1,373	82	5.00	5.97
RP (NonParametric)	1,373	81	5.00	5.90
Max Diversification	1,373	83	5.00	6.05
Equal Weighting	1,373	82	5.00	5.97

Table 18: Historical VaR violations in the out-of-sample period

Portfolio	Observations	Violations	Expected (%)	Observed (%)
RP (Parametric)	1,373	86	5.00	6.26
RP (NonParametric)	1,373	86	5.00	6.26
Max Diversification	1,373	82	5.00	5.97
Equal Weighting	1,373	85	5.00	6.19

Observed violation frequencies range between 5.9% and 6.3% across both methods (Tables 17 and 18), slightly above the theoretical expectation of 5%. The small excess is consistent across all four portfolios and both VaR models, suggesting a common source: the fat-tailed return distribution documented in Section 1.1 causes both Gaussian and historical VaR to modestly underestimate tail risk. Critically, violation rates are nearly identical across strategies, providing no basis for discriminating between them on VaR accuracy grounds. This reflects the high pairwise correlations: when markets move sharply, all four portfolios experience similar return shocks.

3.4 Conclusion

Risk Parity successfully equalizes risk contributions, but when correlations are high it converges toward equal-weighting anyway. The Maximum Diversification portfolio provides the strongest downside protection and the highest Sharpe ratio, making it the preferred choice on risk-adjusted grounds. The Equally Weighted portfolio delivers the highest cumulative return but also the largest drawdowns. Given the homogeneous correlation structure of this technology-heavy universe, the practical differences between strategies are smaller than theory might suggest.

4. VaR of a Bond

4.1 Bond Specification and Conventions

We consider a bond with face value $F = 100$, coupon rate $c = 5\%$ paid annually, maturity $T = 10$ years, and current gross price $P_0 = 99$. The day count convention is 360 days per year. Valuation is assumed to occur immediately following a coupon payment date, implying zero accrued interest at the initial valuation point. Yields are expressed as annual effective rates under annual compounding. Daily changes in the yield to maturity are modelled as i.i.d. Gaussian: $\delta y_{\text{daily}} \sim \mathcal{N}(0, \sigma_y^2)$, where $\sigma_y = 0.006$.

4.2 Bond Pricing and Yield to Maturity

Under the annual compounding convention, the bond price as a function of yield y is:

$$P(y) = \sum_{k=1}^T \frac{CF_k}{(1+y)^k} = C \cdot \frac{1 - (1+y)^{-T}}{y} + F \cdot (1+y)^{-T} \quad (8)$$

where $CF_k = C = 5$ for $k = 1, \dots, 9$ and $CF_{10} = 105$. The price function is strictly monotonically decreasing in yield, since $dP/dy = -\sum_k k CF_k / (1+y)^{k+1} < 0$. This monotonicity is fundamental to the VaR calculation, enabling a direct mapping between yield quantiles and price quantiles. Solving $P(y_0) = 99$ numerically:

$$\boxed{y_0 = 5.1303\%} \quad (9)$$

The bond trades at a slight discount because the coupon rate of 5% lies below the prevailing market yield of 5.1303%.

4.3 Risk Sensitivities

$$\Delta = \frac{\partial P}{\partial y} = -\sum_{k=1}^T \frac{k \cdot CF_k}{(1+y)^{k+1}} = \boxed{-762.47} \quad (10)$$

$$\Gamma = \frac{\partial^2 P}{\partial y^2} = \sum_{k=1}^T \frac{k(k+1) \cdot CF_k}{(1+y)^{k+2}} = \boxed{7,393} \quad (11)$$

$$\Theta = \frac{\partial P}{\partial \tau} \approx \frac{P(y_0, d\tau) - P(y_0, 0)}{d\tau} = \boxed{4.9534 \text{ per annum}} \quad (12)$$

where $d\tau = 1/360$ years. Derived measures: DV01 = 0.07625, modified duration $D_{\text{mod}} = 7.70$ years, Macaulay duration = 8.10 years.

The delta of -762.47 indicates that a 100 bp yield increase causes an approximate price decline of 7.62. The gamma of 7,393 reflects positive convexity: price sensitivity decreases as yields rise, dampening losses under large yield shocks. The theta of 4.9534 per annum represents the pull-to-par effect as the bond approaches maturity.

4.4 Probability of a 10% Price Decline

Over a 30-day horizon, cumulative yield changes follow $\delta y_{30} \sim \mathcal{N}(0, 30\sigma_y^2)$, with $\sigma_{30} = 0.006\sqrt{30} = 0.03286$. A 10% price decline requires $P_{30} \leq 89.10$, equivalent by monotonicity to $y_{30} \geq y^*$ where $P(y^*, 30/360) = 89.10$. Numerical solution gives $y^* = 6.589\%$, a required yield increase of 145.8 basis points:

$$\mathbb{P}(\text{10\% decline}) = 1 - \Phi\left(\frac{0.01458}{0.03286}\right) = 1 - \Phi(0.4437) = 32.86\% \quad (13)$$

Monte Carlo validation with $M = 100,000$ simulations confirms this at 32.90%. The required yield movement of 145.8 bp represents merely 0.44 standard deviations of the 30-day distribution—far from a tail event—illustrating the substantial interest rate risk in long-duration bonds under elevated yield volatility.

4.5 VaR Methodology

$\text{VaR}_{1-\alpha}$ is defined as $-Q_\alpha(\Delta P)$, the loss threshold exceeded only 1% of the time ($\alpha = 0.01$, $z_{0.99} = 2.3263$). Since $P(y)$ decreases monotonically in yield, the left tail of P&L maps directly to the right tail of yield changes. Six methods are implemented.

Method 1: Exact Full Revaluation. Exploits monotonicity: compute $y_{\text{worst}} = y_0 + z_{0.99} \sigma_y \sqrt{n}$, reprice $P_{\text{worst}} = P(y_{\text{worst}}, n/360)$, compute $\text{VaR} = P_0 - P_{\text{worst}}$. This serves as the benchmark, fully capturing the nonlinear price-yield relationship.

Method 2: Delta-Normal. First-order Taylor expansion $\Delta P \approx \Theta\tau + \Delta \delta y$, yielding normally distributed P&L:

$$\text{VaR}^\Delta = -\Theta \frac{n}{360} + |\Delta| z_{0.99} \sigma_y \sqrt{n} \quad (14)$$

Method 3: Delta-Gamma. Second-order expansion: $\Delta P = \mu_\theta + aZ + bZ^2$, where $Z \sim \mathcal{N}(0, 1)$, $a = \Delta\sigma_n$, $b = \frac{1}{2}\Gamma\sigma_n^2$, $\sigma_n = \sigma_y \sqrt{n}$. The quantile is obtained by solving $\Phi(Z_+) - \Phi(Z_-) = 0.01$ via root-finding.

Methods 4–6: Monte Carlo. $M = 10,000$ yield scenarios; P&L computed via delta (MC+ Δ), delta-gamma (MC+ $\Delta\Gamma$), or full repricing (MC+Full). VaR is the empirical α -quantile: $\text{VaR} = -\Delta P_{(k)}$, $k = \lceil M\alpha \rceil = 100$.

4.6 Empirical Results

Table 19 reports 99% VaR across all six methods and ten horizons. Figure 12 plots the four key methods to make the divergence between approaches visually apparent.

Table 19: 99% VaR by Method Across All Horizons

n (days)	Exact	VaR^Δ	$\text{VaR}^{\Delta\Gamma}$	MC+ Δ	MC+ $\Delta\Gamma$	MC+Full
1	9.94	10.63	9.91	10.68	9.95	9.99
10	27.30	33.52	26.32	33.68	26.41	27.41
20	35.57	47.32	32.92	47.55	33.01	35.70
30	40.99	57.88	36.26	58.17	36.35	41.14
40	45.03	66.76	37.82	67.09	37.86	45.17
50	48.21	74.57	38.28	74.94	38.24	48.36
60	50.83	81.61	38.32	82.02	38.32	50.98
70	53.03	88.08	38.26	88.52	38.27	53.18
80	54.92	94.09	38.15	94.56	38.14	55.07
90	56.56	99.73	38.03	100.22	38.02	56.71

At the one-day horizon, the exact VaR of 9.94 represents 10.0% of the bond's dirty price. At 30 days, VaR rises to 40.99, consistent with the 32.86% probability of a 10% decline computed in Section 4.4. The 90-day VaR of 56.56 demonstrates that while losses can be substantial, convexity materially limits downside

relative to linear projections. The delta-gamma VaR exhibits anomalous behaviour, plateauing around 38 and then declining for horizons beyond 50 days — a counter-intuitive result signalling fundamental breakdown of the Taylor approximation, analysed further in Section 4.6.

Table 20: Percentage Errors Relative to Exact VaR

n (days)	VaR^Δ	$\text{VaR}^{\Delta\Gamma}$	$\text{MC}+\Delta$	$\text{MC}+\Delta\Gamma$	$\text{MC}+\text{Full}$
1	+6.9%	−0.3%	+7.4%	+0.1%	+0.5%
10	+22.8%	−3.6%	+23.4%	−3.2%	+0.4%
30	+41.2%	−11.5%	+41.9%	−11.3%	+0.3%
60	+60.6%	−24.6%	+61.4%	−24.6%	+0.3%
90	+ 76.3%	− 32.8%	+77.2%	−32.8%	+0.3%

Both delta-based and delta-gamma methods exhibit large errors at long horizons, but in opposite directions (Table 20), as the divergence in Figure 12 makes clear. Delta-based methods overestimate VaR by up to 77%; delta-gamma methods underestimate by up to 33%. Only MC+Full maintains errors below 0.5% across all horizons.

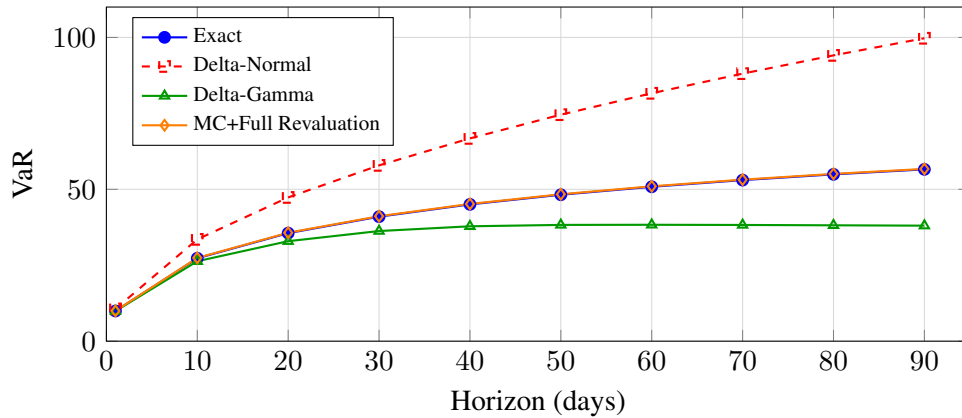


Figure 12: 99% VaR as a function of horizon. Delta-Normal overestimates substantially due to ignoring convexity; Delta-Gamma plateaus and declines beyond 50 days (breakdown analysed in Table 21); only MC+Full accurately tracks the exact solution across all horizons.

4.7 Breakdown of the Delta-Gamma Approximation

At 90 days, the 99th percentile yield shock of 13.24% represents a 258% increase relative to $y_0 = 5.1303\%$. Under such extreme movements, the second-order Taylor approximation overstates the convexity benefit, leading to systematic underestimation of VaR. Table 21 documents how the ratio of the shock to the initial yield grows with horizon and the point at which the approximation breaks down.

Table 21: Taylor Expansion Breakdown Analysis

n (days)	σ_n	99% Δy	$\Delta y/y_0$	Taylor P&L	True P&L
1	0.0060	1.40%	27%	−9.91	−9.94
10	0.0190	4.41%	86%	−26.32	−27.30
30	0.0329	7.65%	149%	−36.27	−40.99
60	0.0465	10.81%	211%	−38.40	−50.83
90	0.0569	13.24%	258%	−34.91	−56.56

The P&L decomposition at the 90-day 99th percentile shock ($\Delta y = 13.24\%$, yields rising from 5.1303% to 18.37%) is reported in Table 22.

Table 22: P&L Component Decomposition at 90 Days (99% Yield Shock)

Component	Contribution	Cumulative
Theta ($\Theta \cdot \tau$)	+1.24	+1.24
Delta ($\Delta \cdot \delta y$)	−100.96	−99.72
Gamma ($\frac{1}{2}\Gamma \cdot \delta y^2$)	+64.82	−34.91
True P&L (Full Repricing)		−56.56

The gamma term of +64.82 drastically overstates the convexity cushion; the corresponding delta-gamma VaR of 38.03 underestimates the exact VaR by 32.8%. When $\Delta y/y_0 > 30\%$, delta-gamma becomes unreliable. For this bond with $\sigma_y = 60$ bp/day, this threshold is exceeded beyond approximately 10–20 days.

4.8 Expected Shortfall

ES at confidence level $1 - \alpha$ is defined as:

$$ES_{1-\alpha} = -\mathbb{E}[\Delta P \mid \Delta P < -\text{VaR}_{1-\alpha}] \quad (15)$$

ES is coherent, satisfying the subadditivity axiom that VaR lacks, and is mandated as the primary market risk measure under Basel IV. Results are reported in Table 23.

Table 23: 99% Expected Shortfall (MC Full Revaluation)

n (days)	VaR (Exact)	ES (MC Full)	ES $^\Delta$	ES/VaR
1	9.94	11.38	12.18	1.145
10	27.30	30.61	38.42	1.121
20	35.57	39.46	54.25	1.109
30	40.99	45.15	66.37	1.101
60	50.83	55.19	93.62	1.086
90	56.56	60.86	114.43	1.076

The ES/VaR ratio of 1.08–1.15 (Table 23) indicates that conditional on breaching the 99% VaR threshold, expected losses exceed VaR by 8–15%. The delta-normal ES at 90 days (114.43) exceeds the bond’s entire price, again demonstrating the breakdown of linear approximations.

A key finding is that **model specification dominates sampling error**. Monte Carlo with a misspecified model converges precisely to an incorrect value; increasing the number of simulations cannot remedy model deficiency. MC+ Δ converges to approximately 58 (overestimate), MC+ $\Delta\Gamma$ converges to approximately 36 (underestimate), while only MC+Full converges to the correct value.

4.9 Conclusion

MC+Full Revaluation is the only reliable method across all horizons, with errors consistently below 0.5%. Delta-Normal overestimates VaR by up to 77% at 90 days, ignoring the convexity cushion. Delta-Gamma underestimates VaR by up to 33%—a potentially dangerous outcome—because the Taylor expansion overstates the convexity benefit when yield shocks are large relative to the initial yield. The 32.86% probability of a 10% price decline within 30 days underscores the substantial interest rate risk in long-duration bonds during periods of elevated yield volatility. All results are conditional on Gaussian yields, constant volatility, and a single-factor parallel shift framework.